IML-Lab -1 — Assignment Writeup

Course: Machine Learning (Semester 5) | **Date:** August 3, 2026 | **Submission Output:** Lab1_writeup.pdf

Practice Question 1: Distance Dynamics & Curse of Dimensionality

Task Statement: For each choice of dimension $d \in \{2^0, 2^1, 2^2, ..., 2^{10}\}$, sample 100 points from the unit cube, and record the average distances between all pairs of points, as well as the standard deviation of the distances.

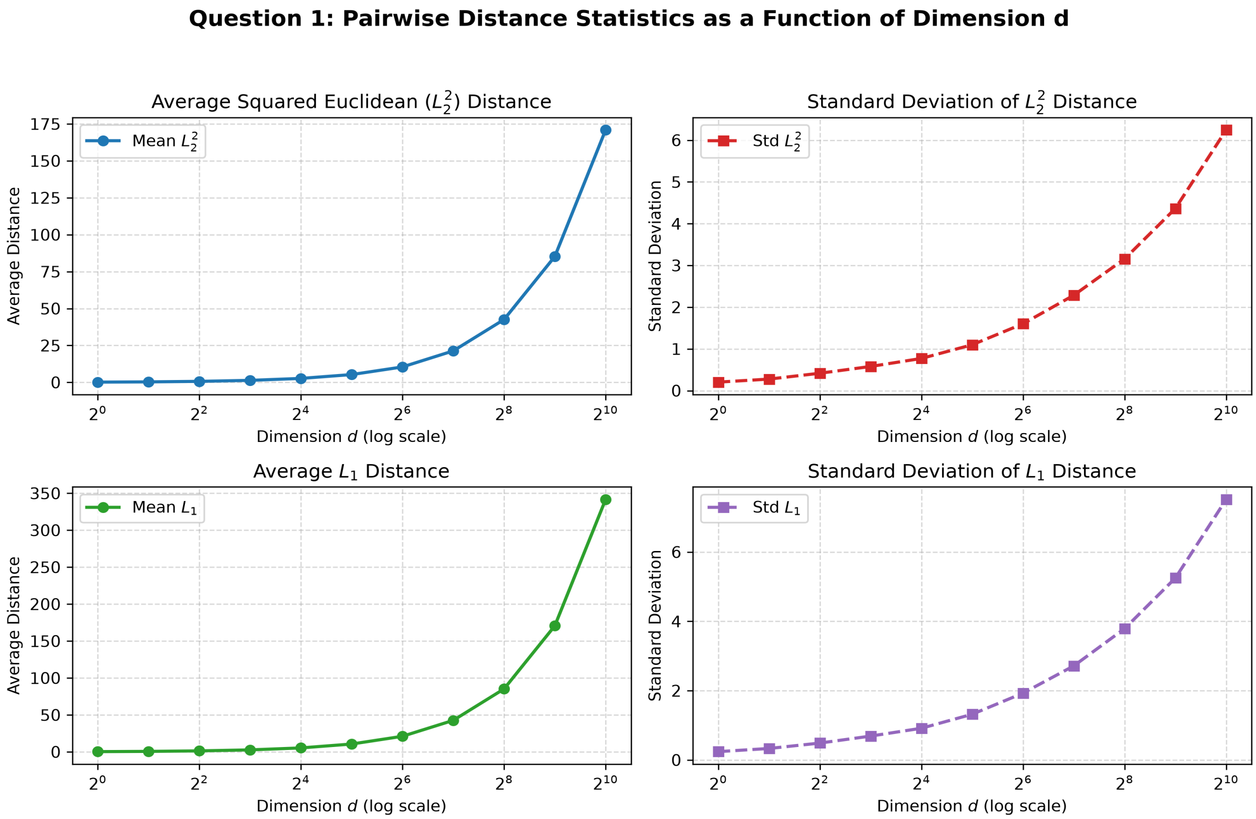
Mathematical Distance Metrics:

i. **Squared Euclidean (L_2^2) distance:** $\|x - y\|_2^2 = \sum_{j=1}^{d} (x_j - y_j)^2$
ii. **L_1 distance (Cityblock):** $\|x - y\|_1 = \sum_{j=1}^{d} |x_j - y_j|$

Recorded Statistics Table:

Dimension (d)	Mean L_2^2	Std L_2^2	Mean L_1	Std L_1
1	0.1770	0.2047	0.3445	0.2416
2	0.3419	0.2773	0.6763	0.3324
4	0.6986	0.4176	1.3661	0.4886
8	1.3678	0.5797	2.6980	0.6900
16	2.6843	0.7720	5.3561	0.9161
32	5.2942	1.0998	10.6254	1.3197
64	10.4584	1.6024	21.0933	1.9290
128	21.3122	2.2863	42.6339	2.7155
256	42.5539	3.1568	85.2097	3.7897
512	85.2249	4.3647	170.6069	5.2494
1024	170.8837	6.2451	341.5763	7.5158

Distance & Standard Deviation Plots as a Function of *d*:



Observations about Distance and Standard Deviation:

- **Linear Growth of Average Distance:** Both mean L_2^2 and mean L_1 distances scale linearly with dimension d (e.g., $E[L_2^2] = d/6$ and $E[L_1] = d/3$).
- **Sub-linear Growth of Standard Deviation:** The standard deviation σ of the pairwise distances grows proportionally to $\sqrt{d}$.
- **Distance Concentration (Curse of Dimensionality):** Because the mean grows linearly with d while the standard deviation grows only as $\sqrt{d}$, the relative variance (σ / μ) approaches zero as $d \to \infty$. Consequently, in high-dimensional spaces (e.g. $d = 1024$), all pairwise distances concentrate around the mean value. Nearest-neighbor distances become nearly equal to farthest-neighbor distances, rendering distance metrics less discriminative.

Practice Question 2: Iris Dataset Workflow & EDA

Objective: Understand the basic workflow of machine learning, including data loading, exploration, preprocessing, and visualization using the benchmark Iris Dataset.

Tasks i – vi: Environment, Dataset Overview & Preprocessing

i. **Libraries Imported:** NumPy, Pandas, Matplotlib, Scikit-learn.
ii. **Dataset Loaded:** Iris dataset from sklearn.datasets.
iii. **DataFrame Conversion:** Dataset converted to a Pandas DataFrame.
v. **Dataset Dimensions:** Number of samples = **150**, Number of features = **4**.
vi. **Missing Values Check: 0 missing values** across all 4 features.

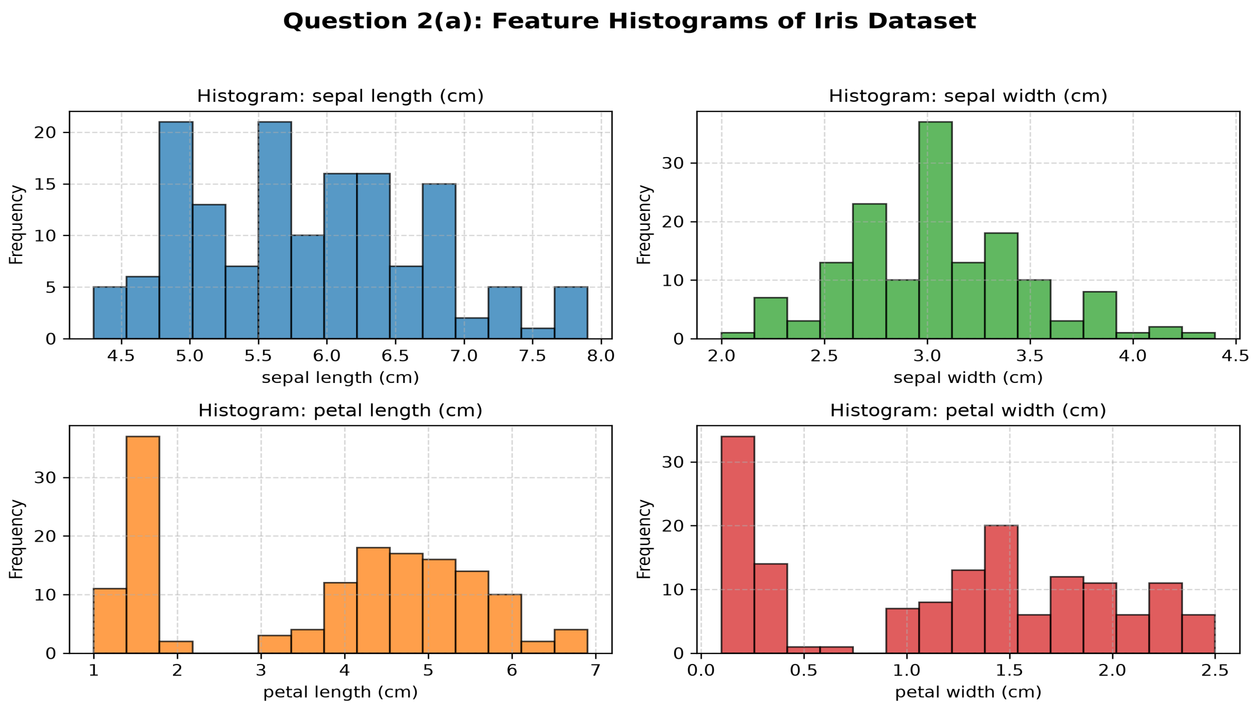
Task iv: Display First 5 Rows of Dataset

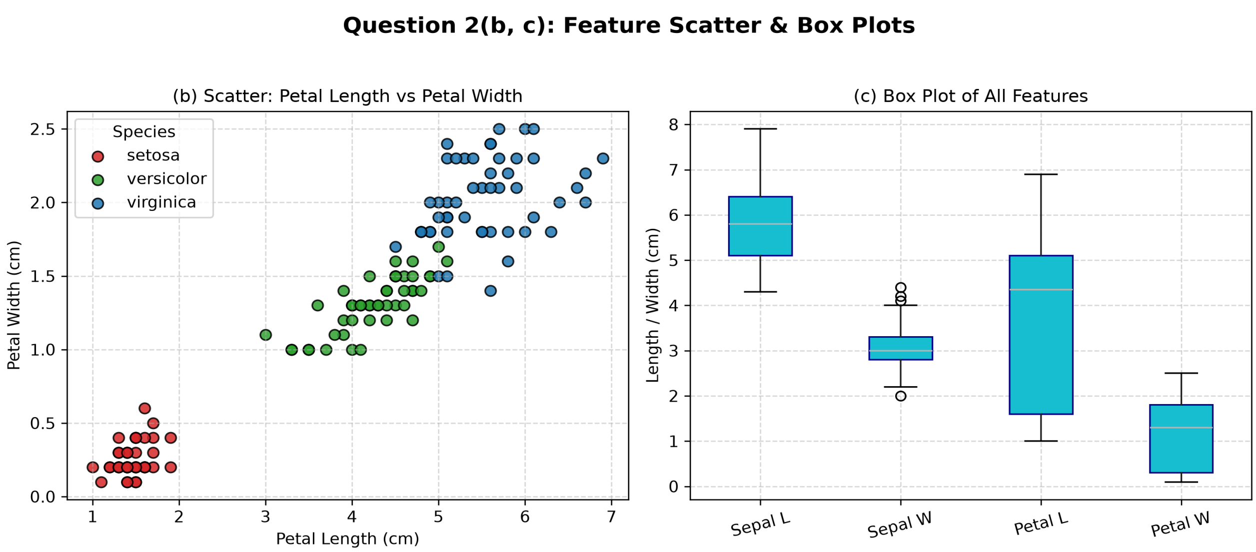
Idx	Sepal L (cm)	Sepal W (cm)	Petal L (cm)	Petal W (cm)	Species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

Task vii: Statistical Measures for Each Feature

Feature Name	Mean	Median	Min Value	Max Value
sepal length (cm)	5.8433	5.8000	4.3000	7.9000
sepal width (cm)	3.0573	3.0000	2.0000	4.4000
petal length (cm)	3.7580	4.3500	1.0000	6.9000
petal width (cm)	1.1993	1.3000	0.1000	2.5000

Task viii: Visualizations (a) Histogram, (b) Scatter Plot, (c) Box Plot





Task ix: Three Key Observations

1. **Linearly Separable Species:** Setosa is easily separable from Versicolor and Virginica based on petal length and width.
2. **High Feature Correlation:** Petal length and petal width display a strong positive linear correlation.
3. **Outlier Presence:** Sepal width exhibits mild outliers on both upper and lower tails, whereas other features show a clean continuous distribution without significant outliers.